Antiderivatives and Riemann Sums

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic antiderivatives

Find the general antiderivative of each function:

a $f(x) = x^4$ $= \frac{x^5}{5} + C$

b $f(x) = \frac{1}{x}$ $= \ln|x| + C$

c $f(x) = e^x$ $= e^x + C$

d $f(x) = \sec^2 x$ $= \tan x + C$

(a) $= \frac{x^5}{5} + C$; (b) $= \ln|x| + C$; (c) $= e^x + C$; (d) $= \tan x + C$

2 [Foundational] • Target: Basic antiderivatives

Find $\int (3x^2 - 4x + 5)\,dx$.

$= x^3 - 2x^2 + 5x + C$.

3 [Foundational] • Target: Basic antiderivatives

Find the particular antiderivative $F(x)$ of $f(x) = 6x^2 - 4$ satisfying $F(1) = 5$.

$F(x) = 2x^3 - 4x + C$. $F(1) = 2 - 4 + C = 5 \Rightarrow C = 7$. $F(x) = 2x^3 - 4x + 7$.

4 [Foundational] • Target: Riemann sums: left, right, and midpoint

Approximate $\int_0^4 x^2\,dx$ using a Left Riemann Sum with $n = 4$ equal subintervals.

$\Delta x = 1$; left endpoints $x = 0, 1, 2, 3$: $L_4 = 1[0^2 + 1^2 + 2^2 + 3^2] = 1[0 + 1 + 4 + 9] = 14$.

5 [Foundational] • Target: Riemann sums: left, right, and midpoint

Approximate $\int_0^4 x^2\,dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

$\Delta x = 1$; right endpoints $x = 1, 2, 3, 4$: $R_4 = 1[1 + 4 + 9 + 16] = 30$.

6 [Foundational] • Target: Riemann sums: left, right, and midpoint

Approximate $\int_0^4 x^2\,dx$ using a Midpoint Riemann Sum with $n = 4$ equal subintervals.

$\Delta x = 1$; midpoints $x = 0.5, 1.5, 2.5, 3.5$: $M_4 = 1[0.25 + 2.25 + 6.25 + 12.25] = 21$. (True value: $\frac{64}{3} \approx 21.33$.)

7 [Foundational] • Target: The Trapezoidal Rule

Use the Trapezoidal Rule with $n = 4$ to approximate $\int_0^4 x^2\,dx$.

$T_4 = \frac{L_4 + R_4}{2} = \frac{14 + 30}{2} = 22$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: The Trapezoidal Rule**

A table gives $f(0) = 2$, $f(2) = 5$, $f(4) = 9$, $f(6) = 8$. Use the Trapezoidal Rule with these unequal-width data (widths of 2 each) to approximate $\int_0^6 f(x)\,dx$.

With 3 equal subintervals of width 2:
$T = 2\left[\frac{f(0)+f(2)}{2} + \frac{f(2)+f(4)}{2} + \frac{f(4)+f(6)}{2}\right] = 2\left[\frac{7}{2} + \frac{14}{2} + \frac{17}{2}\right] = 7 + 14 + 17 = 38.$

9 **[Intermediate]** • **Target: Sums as limits: the definite integral**

Explain how the definite integral $\int_a^b f(x)\,dx$ is defined as a limit of Riemann sums, and write the limit expression using sigma notation.

$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$, where $\Delta x = \frac{b-a}{n}$ and x_i^* is any sample point in the i-th subinterval — as $n \to \infty$ (subintervals shrink to zero width), all Riemann sum types (left, right, midpoint) converge to the same value, the exact signed area under f.

10 **[Intermediate]** • **Target: Sums as limits: the definite integral**

Determine whether the Left Riemann Sum overestimates or underestimates $\int_0^4 x^2\,dx$, and explain why using the shape of the graph.

Since $f(x) = x^2$ is increasing on $[0, 4]$, the left endpoint of each subinterval gives the smallest function value on that subinterval, so the Left Riemann Sum UNDERESTIMATES the true area (confirmed: $L_4 = 14 < \frac{64}{3} \approx 21.33$).

11 **[Intermediate]** • **Target: Riemann sums from a table of values**

A table gives $f(0) = 3$, $f(1) = 5$, $f(2) = 8$, $f(3) = 10$, $f(4) = 9$. Use a Right Riemann Sum with 4 subintervals of width 1 to approximate $\int_0^4 f(x)\,dx$.

$R_4 = 1[f(1) + f(2) + f(3) + f(4)] = 5 + 8 + 10 + 9 = 32.$

12 **[Intermediate]** • **Target: Antiderivatives of trigonometric and exponential functions**

Find $\int (3\sin x + 2e^x)\,dx$.

$= -3\cos x + 2e^x + C.$

13 **[Intermediate]** • **Target: Antiderivatives of trigonometric and exponential functions**

Find the particular antiderivative $F(x)$ of $f(x) = \sec^2 x$ satisfying $F(0) = 2$.

$F(x) = \tan x + C$. $F(0) = 0 + C = 2 \Rightarrow C = 2$. $F(x) = \tan x + 2$.

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14 **[Intermediate]** • **Target: Visualizing a Riemann sum approximation**

The graph shows $f(x) = x^2$ on $[0, 4]$. Use the graph to explain why the Right Riemann Sum $R_4 = 30$ (found earlier) overestimates the true area, and why the Left Riemann Sum $L_4 = 14$ underestimates it.

Since $f(x) = x^2$ is increasing on $[0, 4]$, right-endpoint rectangles reach up to the curve's height at the RIGHT edge of each strip, so their tops poke above the curve, overestimating the area. Left-endpoint rectangles use the height at the LEFT edge (lower, since f is increasing), so their tops stay below the curve, underestimating the area.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: An applied antiderivative problem** • **Cross-Lesson: 6.02 The Fundamental Theorem of Calculus**

A factory's production rate is $r(t) = 100 + 20t$ units/day, where t is days since a process change. Find the total production function $P(t)$ given $P(0) = 0$, then find the total production over the first 10 days.

$P(t) = \int r(t)\,dt = 100t + 10t^2 + C$. Using $P(0) = 0$: $C = 0$. $P(t) = 100t + 10t^2$.
$P(10) = 1000 + 1000 = 2000$ units.

16 **[Challenging]** • **Target: More antiderivative practice** • **Cross-Lesson: 7.01 Differential Equations and Slope Fields**

Find the general antiderivative of $f(x) = 4x^3 - \frac{2}{x^2} + \cos x$.

$= x^4 + \frac{2}{x} + \sin x + C$.

17 **[Challenging]** • **Target: More antiderivative practice** • **Cross-Lesson: 8.01 Area Between Curves**

Find the particular antiderivative $F(x)$ of $f(x) = x^2 - 3$ satisfying $F(0) = 4$.

$F(x) = \frac{x^3}{3} - 3x + C$. $F(0) = 0 - 0 + C = 4 \Rightarrow C = 4$. $F(x) = \frac{x^3}{3} - 3x + 4$.

18 **[Challenging]** • **Target: Comparing Riemann sums to the exact integral** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

For $\int_0^4 x^2\,dx$, the exact value is $\frac{64}{3} \approx 21.33$. Rank the four approximations found earlier — $L_4 = 14$, $R_4 = 30$, $M_4 = 21$, $T_4 = 22$ — from closest to farthest from the exact value, and explain which type of approximation (midpoint or trapezoidal) tends to be more accurate for a smooth curving function.

Distances from 21.33: M_4 ($|21 - 21.33| = 0.33$), T_4 ($|22 - 21.33| = 0.67$), L_4 ($|14 - 21.33| = 7.33$), R_4 ($|30 - 21.33| = 8.67$). Order (closest to farthest): M_4, T_4, L_4, R_4. The midpoint approximation tends to be more accurate than the trapezoidal for smooth curves, because its errors are typically about half the size and opposite in sign to the trapezoidal error.

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19 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 1.01 Limits Graphically and Numerically

Approximate $\int_0^7 5x^2\,dx$ using a Right Riemann Sum with $n = 7$ equal subintervals.

Final Answer: 700. $\Delta x = 1$; right endpoints $x = 1, 2, \ldots, 7$: $R_7 = 5[1^2 + 2^2 + \cdots + 7^2] = 5(140) = 700$.

20 [Challenging] • Target: Antiderivatives of trigonometric and exponential functions • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Approximate $\int_0^6 x^2\,dx$ using a Right Riemann Sum with $n = 6$ equal subintervals.

Final Answer: 91. $\Delta x = 1$; right endpoints $x = 1, 2, \ldots, 6$: $R_6 = 1[1^2 + 2^2 + \cdots + 6^2] = 1(91) = 91$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic antiderivatives

Find C in the particular antiderivative $F(x) = x^3 - 7x + C$ of $f(x) = 3x^2 - 7$, satisfying $F(-1) = -8$.

a. -8 **b.** -12

c. -14 ✓ **d.** -16

$F(-1) = (-1)^3 - 7(-1) + C = 6 + C = -8$, so $C = -14$.

22 [Foundational] • Target: Riemann sums: left, right, and midpoint

Find C in the particular antiderivative $F(x) = -2x^3 - 5x + C$ of $f(x) = -6x^2 - 5$, satisfying $F(2) = -9$.

a. 17 ✓ **b.** -9

c. 15 **d.** 19

$F(2) = -2(2)^3 - 5(2) + C = -26 + C = -9$, so $C = 17$.

23 [Foundational] • Target: The Trapezoidal Rule

Find C in the particular antiderivative $F(x) = 2x^3 + 9x + C$ of $f(x) = 6x^2 + 9$, satisfying $F(0) = 9$.

a. 7 **b.** 9 ✓

c. 10 **d.** 11

$F(0) = 2(0)^3 + 9(0) + C = 0 + C = 9$, so $C = 9$.

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24 [Foundational] • Target: Sums as limits: the definite integral

Find C in the particular antiderivative $F(x) = 2x^3 + x + C$ of $f(x) = 6x^2 + 1$, satisfying $F(3) = -8$.

a. -65 ✓ **b.** -8

c. -67 **d.** -63

$F(3) = 2(3)^3 + (3) + C = 57 + C = -8$, so $C = -65$.

25 [Foundational] • Target: Riemann sums from a table of values

Find C in the particular antiderivative $F(x) = x^3 + 5x + C$ of $f(x) = 3x^2 + 5$, satisfying $F(3) = 4$.

a. 4 **b.** -36

c. -38 ✓ **d.** -40

$F(3) = (3)^3 + 5(3) + C = 42 + C = 4$, so $C = -38$.

26 [Foundational] • Target: Antiderivatives of trigonometric and exponential functions

Find C in the particular antiderivative $F(x) = -2x^3 + 6x + C$ of $f(x) = -6x^2 + 6$, satisfying $F(-3) = -8$.

a. -8 **b.** -46

c. -44 ✓ **d.** -42

$F(-3) = -2(-3)^3 + 6(-3) + C = 36 + C = -8$, so $C = -44$.

27 [Foundational] • Target: Visualizing a Riemann sum approximation

Find C in the particular antiderivative $F(x) = -x^3 + C$ of $f(x) = -3x^2$, satisfying $F(2) = -1$.

a. 9 **b.** 5

c. -1 **d.** 7 ✓

$F(2) = -(2)^3 + C = -8 + C = -1$, so $C = 7$.

28 [Foundational] • Target: An applied antiderivative problem

Approximate $\int_0^4 5x^2\,dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

a. 150 ✓ **b.** 145

c. 170 **d.** 70

$\Delta x = 1$; right endpoints $x = 1, 2, \ldots, 4$: $R_4 = 5[1^2 + 2^2 + \cdots + 4^2] = 5(30) = 150$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: More antiderivative practice

Approximate $\int_0^6 2x^2\,dx$ using a Right Riemann Sum with $n = 6$ equal subintervals.

a. 194

b. 110

c. 180

d. 182 ✓

$\Delta x = 1$; right endpoints $x = 1, 2, \ldots, 6$: $R_6 = 2[1^2 + 2^2 + \cdots + 6^2] = 2(91) = 182$.

30 [Intermediate] • Target: Comparing Riemann sums to the exact integral

Approximate $\int_0^3 5x^2\,dx$ using a Right Riemann Sum with $n = 3$ equal subintervals.

a. 70 ✓

b. 85

c. 65

d. 25

$\Delta x = 1$; right endpoints $x = 1, 2, \ldots, 3$: $R_3 = 5[1^2 + 2^2 + \cdots + 3^2] = 5(14) = 70$.

31 [Intermediate] • Target: Basic Antiderivatives & Power Rule for Integration

Approximate $\int_0^4 3x^2\,dx$ using a Right Riemann Sum with $n = 4$ equal subintervals.

a. 42

b. 90 ✓

c. 102

d. 87

$\Delta x = 1$; right endpoints $x = 1, 2, \ldots, 4$: $R_4 = 3[1^2 + 2^2 + \cdots + 4^2] = 3(30) = 90$.

32 [Intermediate] • Target: Left and Right Riemann Sums

Approximate $\int_0^5 5x^2\,dx$ using a Right Riemann Sum with $n = 5$ equal subintervals.

a. 270

b. 275 ✓

c. 150

d. 300

$\Delta x = 1$; right endpoints $x = 1, 2, \ldots, 5$: $R_5 = 5[1^2 + 2^2 + \cdots + 5^2] = 5(55) = 275$.

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33 [Intermediate] • Target: Midpoint & Trapezoidal Approximations

Approximate $\int_0^5 2x^2\,dx$ using a Right Riemann Sum with $n = 5$ equal subintervals.

a. 110 ✓

b. 120

c. 60

d. 108

$\Delta x = 1$; right endpoints $x = 1, 2, \ldots, 5$: $R_5 = 2[1^2 + 2^2 + \cdots + 5^2] = 2(55) = 110$.

34 [Intermediate] • Target: Limit of a Riemann Sum & Definite Integrals

Approximate $\int_0^6 f(x)\,dx$ using the Trapezoidal Rule (equal width 2), given $f(0) = 7$, $f(2) = 12$, $f(4) = 9$, $f(6) = 3$.

a. 54

b. 52 ✓

c. 62

d. 50

$T = 2\left[\frac{7+12}{2} + \frac{12+9}{2} + \frac{9+3}{2}\right] = 2\left[\frac{7+3}{2} + 12 + 9\right] = 2(5 + 12 + 9) = 52$.

35 [Intermediate] • Target: Basic antiderivatives

Approximate $\int_0^3 f(x)\,dx$ using the Trapezoidal Rule (equal width 1), given $f(0) = 4$, $f(1) = 7$, $f(2) = 12$, $f(3) = 2$.

a. 22 ✓

b. 25

c. 23

d. 21

$T = 1\left[\frac{4+7}{2} + \frac{7+12}{2} + \frac{12+2}{2}\right] = 1\left[\frac{4+2}{2} + 7 + 12\right] = 1(3 + 7 + 12) = 22$.

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36 [Intermediate] • Target: Riemann sums: left, right, and midpoint

For $f(x) = 4x^2$ on $[0, 5]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^5 f(x)\,dx$.

a. Both Left and Right Riemann Sums underestimate.

b. Both Left and Right Riemann Sums overestimate.

c. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.

d. **Left Riemann Sum underestimates, and Right Riemann Sum overestimates.** ✓

$f(x) = 4x^2$ is increasing on $[0, 5]$, so left endpoints give smaller values (underestimate) and right endpoints give larger values (overestimate). So: Left Riemann Sum underestimates, and Right Riemann Sum overestimates.

37 [Intermediate] • Target: The Trapezoidal Rule

For $f(x) = 3x^2$ on $[0, 3]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^3 f(x)\,dx$.

a. Left Riemann Sum overestimates, and Right Riemann Sum underestimates.

b. Both Left and Right Riemann Sums overestimate.

c. **Left Riemann Sum underestimates, and Right Riemann Sum overestimates.** ✓

d. Both Left and Right Riemann Sums underestimate.

$f(x) = 3x^2$ is increasing on $[0, 3]$, so left endpoints give smaller values (underestimate) and right endpoints give larger values (overestimate). So: Left Riemann Sum underestimates, and Right Riemann Sum overestimates.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Sums as limits: the definite integral • Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods

For $f(x) = -2x^2 + 8$ on $[0, 2]$, determine whether the Left and Right Riemann Sums overestimate or underestimate $\int_0^2 f(x)\,dx$.

a. Both Left and Right Riemann Sums underestimate.

b. Both Left and Right Riemann Sums overestimate.

c. **Left Riemann Sum overestimates, and Right Riemann Sum underestimates.** ✓

d. Left Riemann Sum underestimates, and Right Riemann Sum overestimates.

$f(x) = -2x^2 + 8$ is decreasing on $[0, 2]$, so left endpoints give larger values (overestimate) and right endpoints give smaller values (underestimate). So: Left Riemann Sum overestimates, and Right Riemann Sum underestimates.

39 [Challenging] • Target: Riemann sums from a table of values • Cross-Lesson: 1.01 Limits Graphically and Numerically

A factory's production rate is $r(t) = 187 + 10t$ units/day, where t is days since a process change. Find the total production $P(15)$ given $P(0) = 0$.

a. 2805

b. 3920

c. 3930 ✓

d. 3940

$P(t) = \int r(t)\,dt = 187t + 5t^2 + C$. Using $P(0) = 0$: $C = 0$.
$P(15) = 187(15) + 5(15)^2 = 2805 + 1125 = 3930$ units.

40 [Challenging] • Target: Antiderivatives of trigonometric and exponential functions • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

A factory's production rate is $r(t) = 86 + 18t$ units/day, where t is days since a process change. Find the total production $P(14)$ given $P(0) = 0$.

a. 2968 ✓

b. 1204

c. 2950

d. 2986

$P(t) = \int r(t)\,dt = 86t + 9t^2 + C$. Using $P(0) = 0$: $C = 0$.
$P(14) = 86(14) + 9(14)^2 = 1204 + 1764 = 2968$ units.

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41 [Challenging] • Target: Basic Antiderivatives & Power Rule for Integration • Cross-Lesson: 2.01 The Definition of the Derivative

Find $\int (\frac{3}{x} + \frac{1}{x^2})dx$ for $x > 0$.

a. $3\ln x - \frac{1}{x} + C$ ✓

b. $3\ln x + \frac{1}{x} + C$

c. $-\frac{3}{x^2} - \frac{2}{x^3} + C$

d. $3\ln x - \frac{1}{2x} + C$

$\int \frac{3}{x}dx = 3\ln|x|$ and $\int x^{-2}dx = -x^{-1} = -\frac{1}{x}$.

42 [Challenging] • Target: Left and Right Riemann Sums • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Approximate $\int_0^4 x^2 dx$ using a Left Riemann sum with $n = 4$ equal subintervals.

a. 14 ✓

b. 30

c. 21.33

d. 16

$\Delta x = \frac{4-0}{4} = 1$. Subintervals: $[0, 1], [1, 2], [2, 3], [3, 4]$. Left endpoints: $0, 1, 2, 3$. Sum $= 1(0^2 + 1^2 + 2^2 + 3^2) = 1(0 + 1 + 4 + 9) = 14$.

43 [Challenging] • Target: Left and Right Riemann Sums • Cross-Lesson: 3.01 The Chain Rule

If $f(x)$ is strictly increasing on $[a, b]$, what is the relationship between Left and Right Riemann sums and the true integral value I?

a. **Left Sum** $< I <$ **Right Sum** ✓

b. Right Sum $< I <$ Left Sum

c. Left Sum $=$ Right Sum

d. Cannot be determined without concavity

For an increasing function, rectangles using left endpoints underestimate the area while right endpoints overestimate the area: Left $< I <$ Right.

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44 [Challenging] • Target: Midpoint & Trapezoidal Approximations • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

If $f''(x) > 0$ (concave up) on $[a, b]$, how does the Trapezoidal Rule approximation compare to the exact integral?

a. Trapezoidal Rule overestimates the true value ✓

b. Trapezoidal Rule underestimates the true value

c. Trapezoidal Rule is exact

d. Depends on whether f is increasing

Chords connecting points on a concave up curve lie entirely above the curve, so trapezoids overestimate the true area.

45 [Challenging] • Target: Limit of a Riemann Sum & Definite Integrals • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Which definite integral is equal to $\lim_{n\to\infty} \sum_{i=1}^{n} (1 + \frac{2i}{n})^3 \frac{2}{n}$?

a. $\int_1^3 x^3 dx$ ✓

b. $\int_0^2 x^3 dx$

c. $\int_1^2 (1+x)^3 dx$

d. $\int_0^1 (1+2x)^3 dx$

Here $\Delta x = \frac{2}{n}$, interval length $= 2$. $x_i = 1 + i\Delta x \Rightarrow a = 1, b = 3$. The function is $f(x) = x^3$, giving $\int_1^3 x^3 dx$.

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